

Sharp Gaussian Bounds, Cyclic Distribution Torsion, and Extremal Lerch Zeros

Critical Euler constants, exact inverse expansions,
finite jets, and a certified reduction obstruction

A research continuation prepared for the ProveIt project

10 October 2026

Abstract. We continue the manuscript and incoming reports in a fixed ProveIt snapshot. A beta-mixture representation proves strict monotonicity of $-\operatorname{Im} \operatorname{Li}_{a,w-a}(i, 1)$ in the outer order for every positive total order w . This settles the incoming sharp critical Euler conjecture: the best uniform constant is $\pi/4 + (\log 2)/2$. The sharp Gaussian envelope $C(w) = \beta(w) + 2^{-w}\eta(w)$ has a unique nondegenerate maximum, certified at $w = 1.30221658710124\dots$, with value $1.13656110333950\dots$. We prove strict normalized Mellin log-concavity, explicit endpoint identities, Turan inequalities, and an all-order inverse generalized-power expansion with a closed coefficient formula. A positive secant identity represents every Euler remainder for all positive real orders, and proves absolute Euler convergence even below the signed-measure threshold.

For weighted distributions on a finite cyclotomic grid, we calculate coinvariants under a prime-order symmetry with a split fixed factor. The complete integral torsion and all one-variable finite jets are governed by a multiplicative character compatibility condition. For the Stieltjes–Lerch derivative family, a finite resolvent identity moves the largest positive zero strictly left even without saturation. At index two we prove a global lower-branch dichotomy and an exact endpoint criterion; certified signs give a unique minimum at derivative order two and strict decrease of both branches at order three. Finally, an integer functional separates the still-conjectural S_6 target from an enlarged, precisely specified family of 5,131 relations. Analytic proofs, exact replay programs, diagnostic calculations, source provenance, editorial proposals, and further research questions are supplied together.

Baseline commit: `a0a90ef31877f98be437191c48b46f02d5456867`.

Theorems are proved in ordinary mathematics, with exact computation where explicitly indicated. This document is a research contribution for review and integration, not a claim of completed proof-assistant formalization or an exhaustive priority search.

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1 Research baseline and contribution map

The manuscript is a developing research corpus, with newer incoming reports that sometimes settle questions still listed as open elsewhere. A continuation must reconcile that history before selecting targets. We therefore use a single pinned version of Vladimir Reshetnikov’s ProveIt repository [1]: commit `a0a90ef31877f98be437191c48b46f02d5456867`, dated 10 October 2026 UTC. The inspection covered the manuscript’s 40 chapter source files, its editorial and validation material, and all 16 archives present in `docs/incoming`. The 65 retrieved source objects, including those archives, have paths, byte counts, and SHA-256 hashes in the accompanying provenance manifest. This is a source and dependency audit of the present continuation, not an assertion that every proof in the entire corpus has been reverified.

The most useful unresolved targets lie at the intersection of positive integral representations, exact formal relations, and zero geometry. The work below develops all three. Its principal resolved conjecture is the sharp critical constant in the real-order report [2]. The prime-order distribution calculation answers a proposed extension of reflection torsion [3]. The Lerch results extend the branch analysis beyond the cases already proved in the incoming boundary and global-phase reports [15, 16]. The S_6 computation gives a rigorous limitation of a larger finite proof search, while leaving the numerical identity open.

1.1 Conventions and distinctions

We use decreasing-index multiple polylogarithms:

$$\mathrm{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{n_1 > \dots > n_d > 0} \frac{z_1^{n_1} \dots z_d^{n_d}}{n_1^{s_1} \dots n_d^{s_d}},$$

initially in a domain of convergence. In particular $F_{a,b}(z) = \mathrm{Li}_{a,b}(z, 1)$ and $g_{a,b} = \mathrm{Im} F_{a,b}(i)$. The positive real orders a, b need not be integers. At boundary points we specify analytic or Abel values when the ordinary series diverges. We write $\beta(s) = \sum_{j \geq 0} (-1)^j (2j+1)^{-s}$ and $\eta(s) = \sum_{j \geq 1} (-1)^{j-1} j^{-s}$ where their series converge, and use their analytic continuations elsewhere. Standard conventions are also recorded in the DLMF [7, 6].

There are two different weights in this article: $a + b$ is the total order of a Gaussian double polylogarithm, whereas the integer coefficients a_p in a distribution relation are arbitrary formal parameters. The parameters n, k, ρ in the Lerch family denote Stieltjes index, derivative order, and deformation parameter. They are introduced afresh in Section 9.

An equality of complex periods, membership in a specified rational relation span, and a Smith invariant of an integral formal module are different assertions. Our notation and proofs retain those distinctions. In particular, the torsion theorem does not claim torsion among complex numbers, and the S_6 separator does not claim numerical independence.

1.2 What is added to the inspected baseline

The beta allocation identity is the main analytic device. It isolates the split between the two orders in a beta probability law, leaving a strictly decreasing scalar kernel. A covariance then proves monotonicity. Optimizing the endpoint kernel reduces a two-parameter extremum to a one-variable Dirichlet-function problem. Its strict log-concavity is an application of a classical normalized Mellin principle [5], for which we supply the strict proof actually needed.

The distribution calculation follows a different route: a finite integral resolution removes moving orbits from a periodic cohomology complex, and the remaining fixed part becomes a Koszul calculation. The resulting character condition explains how weights on different primes

Topic	Result in this continuation	Scope and proof status
Critical Gaussian values	Strict allocation monotonicity for every positive total order; sharp Euler constant on $a + b = 1$	Analytic proofs; resolves the incoming critical conjecture.
Global envelope	A unique maximum of $\beta(w) + 2^{-w}\eta(w)$, sharp bound for all positive Gaussian orders, and inverse coefficients	Analytic shape theorem plus rational maximum enclosure.
Euler remainders	A positive secant integral and an exact example of increasing normalized errors off the critical line	Analytic identities; prevents an unsupported extension in N .
Cyclic distributions	Integral coinvariants and all one-variable finite jets for a split prime-order action	Complete formula under explicit arithmetic hypotheses.
Lerch zeros	Finite extremal-zero comparison; lower-branch dichotomy; order-two minimum and order-three decrease	Analytic theorems with exact endpoint sign certificates.
Weight-seven reduction	Separation of the S_6 target from 5,131 specified relations through depth four	Integer certificate; the period identity remains conjectural.

can constrain one another. This is more informative than testing each weight separately. Classical universal-distribution work is an essential antecedent, explicitly credited in the theorem section.

For Lerch zeros, positivity acts after a resolvent comparison rather than directly on the original sign-changing kernel. Beyond the largest zero, every shifted summand has a known sign, so a finite comparison works even at parameters where some zeros collide. At index two, the three-block sign pattern of the coefficient sequence additionally restricts all possible stationary points of the lower zero branch.

1.3 Results deliberately retained as prior work

The positive double integral, the sharp finite signed-measure threshold $a + b \geq 1$, and the endpoint atom on $a + b = 1$ are already present in [2]. The current manuscript already proves the S_4 identity; it is not a conjecture to rediscover. Integral freeness, scalar reflection torsion, and reflection jets are already proved in [3]. The index-two Lerch zero count and the unique minimum at derivative order one are also prior results [15]. We recall short arguments where they support new theorems, with explicit attribution.

The references are chosen to support the actual proof mechanisms and to identify nearby literature. They do not establish that every result here is new in the full mathematical literature. The specific continuation claims concern the inspected repository baseline.

2 Allocation of a fixed total order

For positive real orders, define

$$F_{a,b}(z) = \sum_{n \geq 1} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad g_{a,b} = \operatorname{Im} F_{a,b}(i). \quad (2.1)$$

The series initially defines F in the open unit disk. At i we use its principal analytic continuation. This convention matters: if $a + b = 1$, its boundary-series coefficients tend to $1/a$, and the ordinary series does not converge. For integer orders our convention is $F_{a,b}(z) = \text{Li}_{a,b}(z, 1)$, with decreasing summation indices. Write $w = a + b$, and let β and η denote the Dirichlet beta and eta functions. Their continuations are used whenever an ordinary series does not apply.

The positive double integral in the incoming real-order report [2] is the starting point. We recall its short proof to make the subsequent monotonicity argument independent of numerical evidence.

Lemma 2.1 (Positive double integral, prior result). *For $a, b > 0$ and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$F_{a,b}(z) = \frac{z^2}{\Gamma(a)\Gamma(b)} \int_{0 < y < x < 1} \frac{(-\log x)^{a-1} \{\log(x/y)\}^{b-1}}{(1-xz)(1-yz)} dy dx. \quad (2.2)$$

In particular $g_{a,b} < 0$.

Proof. For nonnegative integers j, k , put $y = xv$ and integrate first in v and then in x . The normalized moment of $x^j y^k$ is $(j+k+2)^{-a} (k+1)^{-b}$. Expanding the denominators for $|z| < 1$ therefore reproduces (2.1). The weight has finite mass 2^{-a} , and both denominators stay uniformly away from zero on a compact subset of the slit plane. This also proves holomorphic continuation there. At $z = i$, minus the imaginary part of the rational factor is $(x+y)/\{(1+x^2)(1+y^2)\} > 0$. $\square$

The following change of variables reveals a comparison which is not stated in the inspected reports.

Theorem 2.2 (Strict monotonicity at fixed total order). *For every $w > 0$, the function*

$$a \mapsto -g_{a,w-a}, \quad 0 < a < w,$$

has strictly negative derivative. More precisely, let $V_a \sim \text{Beta}(a, w-a)$ and define

$$J_w(v) = \frac{1}{\Gamma(w)} \int_0^\infty r^{w-1} \frac{e^{-vr} + e^{-r}}{4 \cosh(vr) \cosh r} dr, \quad 0 \leq v \leq 1. \quad (2.3)$$

Then

$$-g_{a,w-a} = \mathbb{E} J_w(V_a), \quad (2.4)$$

$$\frac{d}{da}(-g_{a,w-a}) = \text{Cov}\left(J_w(V_a), \log \frac{V_a}{1-V_a}\right) < 0. \quad (2.5)$$

Its sharp endpoint bounds are

$$\boxed{\beta(w) - \beta(w-1) < -g_{a,w-a} < \frac{\beta(w)}{2} + \frac{\eta(w)}{2^{w+1}}.} \quad (2.6)$$

The lower endpoint is approached as $a \uparrow w$, and the upper endpoint as $a \downarrow 0$.

Proof. In (2.2) put $x = e^{-s}$, $y = e^{-(s+t)}$. The positive integrand for $-g$, including the Jacobian, becomes

$$\frac{s^{a-1} t^{b-1}}{\Gamma(a)\Gamma(b)} \frac{e^{-s} + e^{-(s+t)}}{4 \cosh s \cosh(s+t)} ds dt.$$

Set $r = s + t$ and $v = s/r$. The Jacobian r combines with the powers to give $r^{w-1} v^{a-1} (1-v)^{b-1}$. Dividing by the beta density normalization proves (2.4). Every step is justified by Tonelli's theorem for a positive integrand.

For fixed $r > 0$, the function

$$v \mapsto \frac{e^{-vr} + e^{-r}}{\cosh(vr)}$$

is strictly decreasing. Its numerator decreases strictly and its positive denominator increases. Consequently J_w is strictly decreasing on $[0, 1]$. It is continuous there by domination by the $v = 0$ integrand, which is integrable at both endpoints of the r -axis.

Differentiate the beta density with w fixed. Its score is $\log(v/(1-v)) - \psi(a) + \psi(w-a)$. Beta logarithmic moments are finite for every interior parameter pair, so differentiation under the bounded J_w expectation is legitimate. Centering the score gives (2.5). If V' is an independent copy of V , then

$$2 \operatorname{Cov}(f(V), h(V)) = \mathbb{E}[(f(V) - f(V'))(h(V) - h(V'))].$$

Here $f = J_w$ decreases strictly and $h(v) = \log(v/(1-v))$ increases strictly. The product is strictly negative off the diagonal, which has probability zero. This proves the derivative claim.

Since $\mathbb{E}V_a = a/w$, $V_a \rightarrow 0$ in probability as $a \downarrow 0$; similarly $V_a \rightarrow 1$ as $a \uparrow w$. Boundedness and continuity of J_w prove the asserted endpoint limits. At the first endpoint,

$$\begin{aligned} J_w(0) &= \frac{1}{\Gamma(w)} \int_0^\infty r^{w-1} \frac{1 + e^{-r}}{4 \cosh r} dr \\ &= \frac{1}{2} \beta(w) + 2^{-w-1} \eta(w). \end{aligned} \tag{2.7}$$

The beta and eta Mellin integrals converge for $w > 0$; their series identifications follow first where absolute convergence holds and then through those integrals. For the other endpoint, put $q(r) = (2 \cosh r)^{-1}$. One has

$$\frac{e^{-r}}{2 \cosh^2 r} = q(r) + q'(r).$$

Integration by parts for $w > 1$ gives $J_w(1) = \beta(w) - \beta(w-1)$. Both sides are holomorphic for $\operatorname{Re} w > 0$, so the same equality holds for all positive w . For $0 < w \leq 1$ it is an analytic-continuation identity, not a manipulation of a convergent series for $\beta(w-1)$. $\square$

This theorem concerns a redistribution of total order: increasing a simultaneously decreases b . It is different from a comparison in one index while the other remains fixed. The proof works below, on, and above the finite signed-measure threshold $a + b = 1$.

3 The critical Euler conjecture is a theorem

The incoming report *The Sharp Real-Order Threshold for Harmonic Polylogarithms* labels strict decrease of $-2g_{a,1-a}$ and its proposed uniform Euler constant as Conjecture `conj:constant` [2]. Theorem 2.2 settles its monotonicity part. We spell out the error implication because ordinary convergence and the normalization of the remainder must be kept distinct.

For $a + b = 1$ set

$$f_m = \frac{H_{2m}^{(b)}}{(2m+1)^a}, \quad d_k = \sum_{j=0}^k (-1)^j \binom{k}{j} f_j, \quad E_N = \sum_{k=0}^{N-1} \frac{d_k}{2^{k+1}}. \tag{3.1}$$

The forward difference used here is $\Delta f_m = f_m - f_{m+1}$.

Lemma 3.1 (Critical endpoint measure, prior result). *For $0 < a < 1$, $b = 1 - a$, there is a finite positive measure $\nu_{a,b}$ on $(0, 1)$, with strictly positive density and total mass $1/a$, such that*

$$\frac{H_{n-1}^{(b)}}{n^a} = \frac{1}{a} - \int_0^1 u^{n-1} d\nu_{a,b}(u). \tag{3.2}$$

Equivalently, the signed representing measure is $a^{-1}\delta_1 - \nu_{a,b}$.

Proof. We recall the construction from [2]. Write $k_s(T) = T^{s-1}/\Gamma(s)$ and

$$h(t) = \frac{1}{1-e^{-t}} - \frac{1}{t}, \quad C_{a,b}(T) = \int_0^T k_a(T-t)k_b(t)h(t) dt.$$

The regularized Hurwitz integral [6] gives

$$\int_0^\infty e^{-nT} C_{a,b}(T) dT = n^{-a} \left(\zeta(b, n) - \frac{n^{1-b}}{b-1} \right).$$

Since $b = 1 - a$, the last term contributes $1/a$. Define

$$d\nu_{a,b}(u) = \{-\zeta(b)k_a(-\log u) + C_{a,b}(-\log u)\} du.$$

Both summands are positive: $\zeta(b) < 0$ for $0 < b < 1$ and $h > 0$. The beta integral gives $0 < C_{a,b}(T) < k_{a+b}(T) = 1$. Thus the density is integrable, and its $(n-1)$ st moment is $1/a - n^{-a}(\zeta(b) - \zeta(b, n))$. The finite Hurwitz difference is $H_{n-1}^{(b)}$. Taking $n = 1$ proves its mass and the formula. $\square$

Theorem 3.2 (Sharp critical-line Euler constant). *For every $0 < a < 1$, $b = 1 - a$, and integer $N \geq 1$,*

$$0 < E_N - g_{a,b} < \left(\frac{\pi}{4} + \frac{\log 2}{2} \right) 2^{-N}. \quad (3.3)$$

The displayed constant is optimal simultaneously over this entire critical line and all $N \geq 1$. For fixed a , the exact optimal non-strict constant is $-2g_{a,1-a}$, attained at $N = 1$, and

$$\frac{\pi}{2} - 1 < -2g_{a,1-a} < \frac{\pi}{4} + \frac{\log 2}{2}. \quad (3.4)$$

The normalized errors $2^N(E_N - g_{a,b})$ decrease strictly to zero as $N \rightarrow \infty$.

Proof. Equation (3.2) implies $d_0 = 0$ and, for $k \geq 1$,

$$d_k = - \int_0^1 (1-u^2)^k d\nu_{a,b}(u) < 0.$$

The finite measure justifies summing the absolutely convergent Euler series under the integral. The analytic value at i is its sum, and the geometric tail is

$$2^N(E_N - g_{a,b}) = \int_0^1 \frac{(1-u^2)^N}{1+u^2} d\nu_{a,b}(u). \quad (3.5)$$

The integrand decreases strictly with N for $0 < u < 1$ and converges to zero. Its value at $N = 1$ integrates to $-2g_{a,b}$ because $E_1 = 0$. This proves the fixed-parameter statement and strict convergence. Theorem 2.2 at $w = 1$ gives (3.4), using $\beta(1) = \pi/4$, $\beta(0) = 1/2$, and $\eta(1) = \log 2$. It also gives the strict uniform upper bound in (3.3). Finally, take $N = 1$ and $a \downarrow 0$ to show that no smaller uniform constant can work. $\square$

The atom at $u = 1$ is essential in this argument. The ordinary terms $(-1)^m f_m$ do not approach zero on the critical line; Euler acceleration computes the analytic or Abel value. No divergent series has been silently declared ordinarily convergent.

4 A sharp global envelope and its unique maximum

Define

$$C(w) = \beta(w) + 2^{-w}\eta(w), \quad D(w) = C(w) - 1 \quad (w > 0). \quad (4.1)$$

Theorem 2.2 says that $C(w)$ is the sharp upper envelope of $-2g_{a,b}$ at fixed $a + b = w$. The next theorem determines its global shape. The normalized Mellin log-concavity principle used in the proof is classical; see, for example, Proposition 3.2 of Fradelizi–Li–Madiman [5]. We provide a strict version with a direct proof rather than making a new priority claim for that principle.

Lemma 4.1 (Strict normalized Mellin log-concavity). *Suppose $f : (0, \infty) \rightarrow (0, \infty)$ is strictly log-concave, all the Mellin integrals and their first two parameter derivatives below are finite, and these derivatives can be passed under the integral. Then*

$$M(p) = \frac{1}{\Gamma(p)} \int_0^\infty r^{p-1} f(r) dr$$

has $(\log M)''(p) < 0$ for every $p > 0$ in that domain.

Proof. Normalize $r^{p-1}f(r)$ to a probability density P . Choose a gamma density Q of shape p and rate λ with

$$\mathbb{E}_Q \log r = \psi(p) - \log \lambda = \mathbb{E}_P \log r.$$

Both $\int(P - Q)$ and $\int \log r(P - Q)$ vanish. The logarithm of the likelihood ratio is $\log f(r) + \lambda r +$ a constant, hence is strictly concave. Its positive set is an interval. Normalization forces both signs. A single crossing is impossible: multiplying the difference by $\log(r/r_0)$ at that crossing would give a nonzero quantity of one sign, contrary to the two matching moments. Thus there are two crossings $r_1 < r_2$ and the sign pattern of $P - Q$ is negative, positive, negative.

Consequently

$$\int_0^\infty (\log r - \log r_1)(\log r - \log r_2)(P - Q)(r) dr < 0.$$

The constant and linear terms disappear by the matching moments. Therefore $\text{Var}_P(\log r) < \text{Var}_Q(\log r) = \psi_1(p)$. Differentiation of $\log M$ identifies its second derivative with precisely the difference of these two variances. This proves the lemma. $\square$

Theorem 4.2 (Unique maximum of the sharp Gaussian envelope). *For every $w > 0$, $D(w) > 0$ and $(\log D)''(w) < 0$. There is a unique $w_* > 0$ where C attains its global maximum $C_* = C(w_*)$. It satisfies $1 < w_* < 2$ and the following certified enclosures:*

$$1.30221658710124120959237170 < w_* < 1.30221658710124120959237171. \quad (4.2)$$

Writing $L_C < C_* < U_C$, the certified value endpoints are

$$\begin{aligned} L_C &= 1.136561103339509560952586375779942387681723540, \\ U_C &= 1.136561103339509560952586375779942387681723541. \end{aligned} \quad (4.3)$$

In particular

$$0 < -2 \operatorname{Im} F_{a,b}(i) < C_* \quad (a, b > 0), \quad (4.4)$$

and this global constant is optimal, approached with $a \downarrow 0$ and $b = w_* - a$.

Proof. Subtract the gamma integral for 1 from (2.7). One obtains

$$D(w) = \frac{1}{\Gamma(w)} \int_0^\infty r^{w-1} f(r) dr, \quad f(r) = \frac{e^{-r}(1 - e^{-r})}{2 \cosh r} > 0. \quad (4.5)$$

The logarithm of this density has

$$(\log f)''(r) = -\frac{e^r}{(e^r - 1)^2} - \operatorname{sech}^2 r < 0. \quad (4.6)$$

Near zero $f(r) = r/2 + O(r^2)$, and at infinity it decays exponentially. Thus all the conditions of Lemma 4.1 hold on $w > 0$. This proves strict log-concavity and positivity.

The integral of $f(r)/r$ is finite, so $D(w) \rightarrow 0$ as $w \downarrow 0$. Also $f(r) < e^{-2r}$, whence $D(w) < 2^{-w} \rightarrow 0$ at infinity. Positivity, continuity, and these endpoint limits ensure an interior maximum. Strict concavity of $\log D$ makes that maximum unique and nondegenerate. The exact certificate in Section 6 proves $C'(1) > 0$, $C'(2) < 0$, and the sharper derivative signs at the two rational endpoints in (4.2). Strict log-concavity then locates the unique critical point. The same certificate uses a tangent upper bound for $\log D$ to enclose its value as in (4.3). Finally (4.4) and optimality follow from the sharp allocation endpoints in Theorem 2.2. $\square$

Scope of the constant. The number C_* is a bound for $-2g$, equivalently for the normalized *first* Euler remainder. This theorem does not assert that the same constant bounds $2^N(E_N - g)$ for every N throughout $a + b \geq 1$. The all- N optimal constant proved here is the critical-line constant in Theorem 3.2. Off that line, normalized errors need not decrease with N .

Corollary 4.3 (Turan inequalities and a second normalization). *For $w > 0$ and $0 < h < w$,*

$$D(w)^2 > D(w - h)D(w + h). \quad (4.7)$$

Moreover $2^w D(w)$ increases strictly from zero to one on $(0, \infty)$. For example,

$$\begin{aligned} \left(G + \frac{\pi^2}{48} - 1\right)^2 &> \left(\frac{\pi}{4} + \frac{\log 2}{2} - 1\right) \\ &\quad \cdot \left(\frac{\pi^3}{32} + \frac{3\zeta(3)}{32} - 1\right), \end{aligned} \quad (4.8)$$

where $G = \beta(2)$ is Catalan's constant.

Proof. The first assertion is strict midpoint log-concavity. For the second, rewrite (4.5) using a gamma random variable R_w of shape w and rate 2:

$$2^w D(w) = \mathbb{E} \frac{1 - e^{-R_w}}{1 + e^{-2R_w}}.$$

The function of R_w is strictly increasing from zero to one. If $w_2 > w_1$, couple $R_{w_2} = R_{w_1} + Y$ with an independent positive gamma variable Y of shape $w_2 - w_1$ and rate 2. This gives strict increase. The limit at zero follows by the gamma mean, and the limit at infinity by concentration or the gamma Laplace transform. The displayed constant inequality is (4.7) at $w = 2, h = 1$. $\square$

5 All-order inversion of the envelope

There is a unique solution $W(x) > w_*$ of $C(W(x)) = 1 + x$ for $0 < x < C_* - 1$. Its large-order expansion is a concrete inverse transseries for a combination of Dirichlet L -values. We include an all-coefficient formula, since simply replacing $D(w)$ by 2^{-w} misses several noninteger scales.

For $n \geq 3$, put

$$\epsilon_n = \begin{cases} 1, & n \equiv 1, 2 \pmod{4}, \\ -1, & n \equiv 0, 3 \pmod{4}, \end{cases} \quad \lambda_n = \log_2(n/2) > 0. \quad (5.1)$$

For a finite-support multi-index $\mathbf{k} = (k_3, k_4, \dots)$ of nonnegative integers, write $K = \sum k_n$ and $\Lambda(\mathbf{k}) = \sum k_n \lambda_n$. Set $(u)_0 = 1$ and $(u)_j = u(u+1) \cdots (u+j-1)$.

Theorem 5.1 (Explicit inverse generalized-power expansion). *For every fixed $A > 0$, as $x \downarrow 0$,*

$$W(x) = \frac{\log(1/x)}{\log 2} + \frac{1}{\log 2} \sum_{\substack{\mathbf{k} \neq 0 \\ \Lambda(\mathbf{k}) < A}} \frac{(-1)^{K+1} (\Lambda(\mathbf{k}) + 1)_{K-1}}{\prod_n k_n!} \left(\prod_n \epsilon_n^{k_n} \right) x^{\Lambda(\mathbf{k})} + O(x^A). \quad (5.2)$$

The sum is finite, and coefficients at coincident exponents are added. Writing $\lambda = \log_2(3/2)$ and $\mu = \log_2(5/2)$, its first terms are

$$W(x) = \frac{\log(1/x)}{\log 2} - \frac{x^\lambda + x}{\log 2} - \frac{\lambda + 1/2}{\log 2} x^{2\lambda} + \frac{x^\mu}{\log 2} - \frac{\lambda + 1}{\log 2} x^{\lambda+1} - \frac{(3\lambda + 1)(3\lambda + 2)}{6 \log 2} x^{3\lambda} + O(x^{\log_2(7/2)}). \quad (5.3)$$

Proof. For $w > 1$ the absolutely convergent Dirichlet series gives

$$D(w) = 2^{-w} + \sum_{n \geq 3} \epsilon_n n^{-w}.$$

Put $y = 2^{-w}$ and $y = xt$. Then $D(w) = x$ is equivalent to

$$t + \sum_{n \geq 3} \epsilon_n x^{\lambda_n} t^{1+\lambda_n} = 1. \quad (5.4)$$

For any fixed target order A , keep finitely many n with $\lambda_n \leq A$ and regard the x^{λ_n} as independent small variables. The implicit derivative in t at the origin is 1. The ordinary analytic implicit-function theorem therefore applies.

To justify discarding the rest, for a fixed sufficiently large N and $w \rightarrow \infty$ the integral test gives

$$\sum_{n \geq N} (2/n)^w \leq (2/N)^w \left(1 + \frac{N}{w-1} \right).$$

Also $D(w) \sim 2^{-w}$, so $y/x \rightarrow 1$. Taking $\lambda_N > A$ makes the discarded part $o(x^A)$, uniformly for t in a fixed neighborhood of 1. Stability of the implicit solution preserves that error. Since every $\lambda_n \geq \lambda_3 > 0$ and only finitely many are below a fixed bound, the additive exponent semigroup is locally finite. Taylor truncation hence gives an asymptotic expansion to every fixed order.

For the coefficients, introduce a scalar bookkeeping parameter z and write $t = 1 - z \sum_n \epsilon_n X_n t^{1+\lambda_n}$. Lagrange inversion applied to $\log t$ gives, for $K \geq 1$,

$$[X^{\mathbf{k}}] \log t = \frac{(-1)^K (K-1)!}{\prod k_n!} \left(\prod \epsilon_n^{k_n} \right) \binom{K + \Lambda - 1}{K-1}.$$

Now $W(x) = \log(1/x)/\log 2 - \log t/\log 2$, and $(K-1)! \binom{K+\Lambda-1}{K-1} = (\Lambda+1)_{K-1}$. This proves (5.2). In (5.3), the exponent $\lambda + 1 = \lambda_6$ receives both the direct $n = 6$ term and the product of the $n = 3$ and $n = 4$ terms. Their combined coefficient is $-(\lambda + 1)/\log 2$. This collision is why terms must be grouped before an expansion is reported. $\square$

6 Exact enclosure of the Gaussian maximum

The script `code/certify_gaussian_envelope.py` uses only integer and rational arithmetic. A decimal grid with 90 digits is an implementation choice for outward rounding, not an assumption

about the accuracy of a transcendental library. We give the error bounds which connect its finite checks to Theorem 4.2.

For $d = 1, 2$ let

$$A_d(w) = \sum_{j \geq 0} \frac{(-1)^j}{(dj + 1)^w}, \quad a_j(w) = (dj + 1)^{-w}.$$

Thus $A_1 = \eta$, $A_2 = \beta$. Their Euler partial sums are

$$T_{d,N}(w) = 2^{-N} \sum_{j=0}^{N-1} (-1)^j (dj + 1)^{-w} \sum_{h=j+1}^N \binom{N}{h}. \quad (6.1)$$

With $T \sim \text{Gamma}(w, 1)$, the exact tail is

$$A_d(w) - T_{d,N}(w) = 2^{-N} \mathbb{E} \frac{(1 - e^{-dT})^N}{1 + e^{-dT}}. \quad (6.2)$$

The gamma integral proves this identity first by a geometric expansion; the bounded expectation then justifies it for every $w > 0$. It is positive and at most 2^{-N} .

Lemma 6.1 (Differentiated Euler enclosure). *For $w \geq 1$,*

$$|A'_d(w) - T'_{d,N}(w)| < 2^{1-N}.$$

Consequently, if $C_N = T_{2,N} + 2^{-w}T_{1,N}$, then

$$0 < C(w) - C_N(w) \leq (1 + 2^{-w})2^{-N}, \quad |C'(w) - C'_N(w)| < 4 \cdot 2^{-N}. \quad (6.3)$$

Proof. Differentiating the gamma density in (6.2) introduces $\log T - \psi(w)$. Cauchy–Schwarz bounds its absolute mean by $\sqrt{\psi_1(w)} \leq \sqrt{\psi_1(1)} = \pi/\sqrt{6} < 2$. This proves the first bound. The product rule for $2^{-w}A_1$ gives the derivative bound $2^{-N}\{2(1 + 2^{-w}) + 2^{-w} \log 2\} < 4 \cdot 2^{-N}$. All differentiated integrals are dominated on compact w -intervals. $\square$

The finite centers in (6.1) require logarithms and powers at rational w . The verifier bounds $\log n$ by reducing to $1 \leq r \leq 2$ and using

$$\log r = 2 \sum_{j=0}^{M-1} \frac{v^{2j+1}}{2j+1} + R_M, \quad v = \frac{r-1}{r+1}, \quad 0 \leq R_M \leq \frac{2v^{2M+1}}{(2M+1)(1-v^2)}.$$

It bounds exponentials after scaling their argument to $[0, 1]$, using the positive Taylor series and a geometric tail, followed by outward reciprocals and four squarings. Thus $(dj + 1)^{-w}$ and $-\log(dj + 1)(dj + 1)^{-w}$ both have rational enclosing intervals. Every addition, multiplication and division rounds outward.

With $N = 160$, the program proves positive and negative derivative signs at the rational endpoints of (4.2). It separately proves

$$C'(1) > 0.03260, \quad C'(2) < -0.03561.$$

To enclose the maximum value, let $\ell < u$ be the certified endpoints. Strict log-concavity gives, for $\ell \leq w \leq u$,

$$D(w) \leq D(\ell) \exp\left(\frac{C'(\ell)}{D(\ell)}(u - \ell)\right).$$

Outward bounds for the right-hand side and the lower bound $C(\ell)$ give (4.3). This is a rational certificate for the maximum, not a floating-point optimizer followed by rounded digits.

Independent high-precision quadrature evaluates the beta-mixture and Mellin integrals. In particular it checks the derivative at $w = 1$ without differentiating the removable eta singularity numerically. These diagnostics are useful checks on conventions, but the rational certificate and the preceding analytic proofs supply the stated bounds.

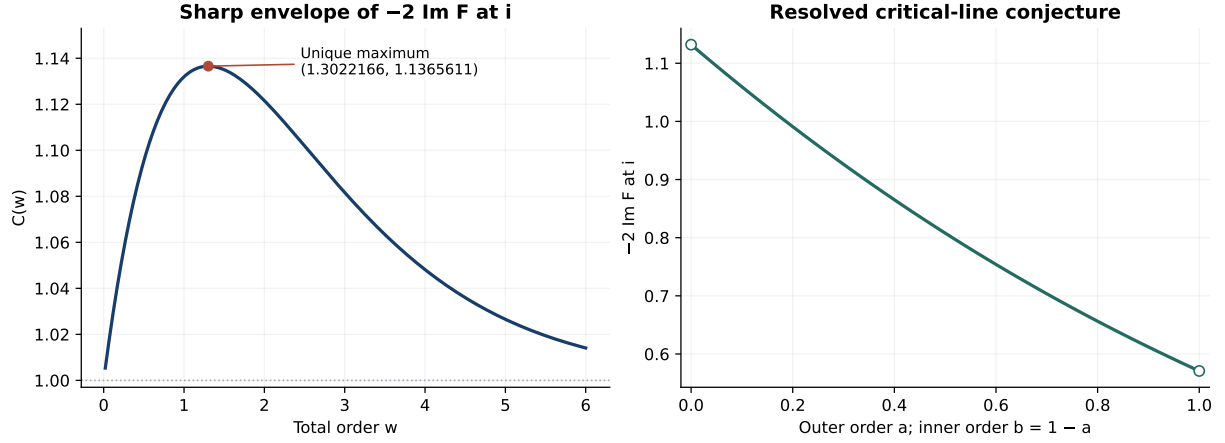


Figure 1: The sharp Gaussian envelope and its critical-line allocation. The curves are diagnostic plots; the shapes and the maximum enclosure are established by Theorems 2.2–4.2. Open endpoint markers denote limiting values, not allowed parameter pairs.

7 A positive secant kernel for every Euler remainder

Let $a, b > 0$, $w = a + b > 0$, and let $R_N = 2^N(E_N - g_{a,b})$ use the Euler partial sums defined in (3.1), now for arbitrary $a, b > 0$. Define

$$H_N(s) = \frac{(1 - s^{-2})^N}{1 + s^{-2}}, \quad s \geq 1.$$

Theorem 7.1 (Positive secant identity). *For every $a, b > 0$, the Euler series converges absolutely to the analytic value $g_{a,b}$. Its partial sums decrease strictly to that value. For every integer $N \geq 1$, with $V \sim \text{Beta}(a, b)$,*

$$R_N = \mathbb{E} \left[\frac{1}{\Gamma(w)} \int_0^\infty r^{w-1} \frac{H_N(e^r) - H_N(e^{Vr})}{e^r - e^{Vr}} dr \right]. \quad (7.1)$$

The quotient is positive for $0 < V < 1$. Its extension to $V = 1$ is $H'_N(e^r)$. Formula (7.1) gives a positive integral. When a finite signed Hausdorff representation exists, its formula for the same remainder instead uses cancellation between positive and negative parts. The positive secant formula also works below the signed-measure threshold.

Proof. Start with the positive Mellin representation, where $f(n) = f_n$,

$$f(n) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^\infty \int_0^\infty u^{a-1} t^{b-1} \frac{e^{-u}}{e^t - 1} (e^{-2nu} - e^{-2n(u+t)}) du dt.$$

This is the finite geometric-sum formula for $H_{2n}^{(b)}$ followed by the gamma integral for $(2n+1)^{-a}$. Applying the finite binomial transform gives

$$-d_k = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^\infty \int_0^\infty u^{a-1} t^{b-1} \frac{e^{-u}}{e^t - 1} \left((1 - e^{-2(u+t)})^k - (1 - e^{-2u})^k \right) du dt.$$

Every displayed integrand is nonnegative. Tonelli's theorem therefore permits summing $\sum_{k \geq N} 2^{N-k-1}(-d_k)$ under the double integral. Summing the two geometric tails produces

$$\frac{e^{-u}}{e^t - 1} (H_N(e^{u+t}) - H_N(e^u)) = \frac{H_N(e^{u+t}) - H_N(e^u)}{e^{u+t} - e^u}.$$

The beta change of variables $r = u + t$, $v = u/r$ gives the asserted integral for this nonnegative series. To establish its finiteness and identify the Euler sum without a restriction on w , first take $N = 1$. Direct simplification shows that its secant kernel is twice the kernel in (2.3). Therefore

$$\sum_{k \geq 1} \frac{-d_k}{2^{k+1}} = \mathbb{E}J_w(V) = -g_{a,b} < \infty.$$

This proves absolute convergence to the analytic value for every positive pair. Also $d_k < 0$ for $k \geq 1$, since its displayed positive integrand is strictly positive. Hence the partial sums decrease strictly. Every later geometric tail is finite and equals $2^N(E_N - g_{a,b})$, proving the formula for all $N \geq 1$. No conditionally convergent rearrangement is needed. $\square$

For $N = 1$, $H_1(s) = (s^2 - 1)/(s^2 + 1)$ and

$$H_1''(s) = \frac{4(1 - 3s^2)}{(1 + s^2)^3} < 0 \quad (s \geq 1).$$

Consequently its secant slope decreases as the lower endpoint e^{vr} increases. Direct simplification gives

$$\frac{H_1(e^r) - H_1(e^{vr})}{e^r - e^{vr}} = \frac{e^{-vr} + e^{-r}}{2 \cosh(vr) \cosh r},$$

which is exactly twice the kernel used in the allocation theorem. For $N = 2$, however, $H_2(1+h) = 2h^2 + O(h^3)$, so H_2 is convex near one. The same pointwise concavity proof does not extend to every N .

There is a particularly small exact obstruction to monotonicity in Euler order away from the critical line. At $(a, b) = (1, 1)$,

$$g_{1,1} = \operatorname{Im} \frac{1}{2} \log^2(1 - i) = -\frac{\pi \log 2}{8}, \quad f(1) = \frac{H_2}{3} = \frac{1}{2}.$$

Since $E_1 = 0$ and $E_2 = -f(1)/4 = -1/8$,

$$R_1 = \frac{\pi \log 2}{4}, \quad R_2 = \frac{\pi \log 2}{2} - \frac{1}{2}, \quad R_2 - R_1 = \frac{\pi \log 2 - 2}{4} > 0.$$

The last inequality follows already from $\pi > 3$ and $\log 2 = 2 \operatorname{artanh}(1/3) > 2/3$. This does not challenge the critical-line monotonicity theorem, whose endpoint atom supplies a different argument. It shows why the sharp bound for $-2g$ and the first Euler remainder cannot be declared the sharp bound for every Euler order without an additional argument.

8 Cyclic prime symmetries and integral torsion

The incoming integral-distribution reports [3] determine reflection torsion and explicitly propose a subgroup of prime order as a next target. We solve that problem for an action which is trivial on one primary factor and has no nonzero fixed point on the complementary factor, assuming the fixed factor has semisimple unit-group action modulo the symmetry prime. The result includes arbitrary integral weights and all one-variable integral jets. Its obstruction is a finite multiplicative compatibility condition, rather than a condition on each weight in isolation.

The finite resolution and double-complex method have classical antecedents in Kubert, Anderson, and Ouyang [10, 11, 12]. In particular, Ouyang's universal norm distributions already provide a much broader setting for weighted distribution relations and their cohomology. The calculation below is an explicit continuation for the specified finite grid and a diagonal cyclic subgroup. We claim no global priority for its ordinary-weight specialization or for the resolution method.

8.1 The formal module and the hypotheses

For a commutative ring B , let $X_q = (1/q)\mathbb{Z}/\mathbb{Z}$ and choose weights $a_p \in B$, one for each prime $p \mid q$. Define

$$D_q(B, \mathbf{a}) = \frac{B[X_q]}{\left\langle \sum_{py=x} e_y - a_p e_x : p \mid q, x \in X_{q/p} \right\rangle}. \quad (8.1)$$

Every root sum is taken in X_q . The symbol e_0 is retained. The integral normal-form theorem from the incoming reports, recalled below with its proof mechanism, gives a B -free module of rank $\varphi(q)$ for every specialization.

Let ℓ be a prime and let $u \in (\mathbb{Z}/q\mathbb{Z})^\times$ have order ℓ . Multiplication by u induces $T(e_x) = e_{ux}$. We impose the following explicit hypotheses:

$$q = md, \quad (m, d) = 1, \quad d > 1, \quad u \equiv 1 \pmod{m}, \quad (u-1, d) = 1, \quad \ell \nmid \varphi(m). \quad (8.2)$$

Here $\varphi(1) = 1$. Necessarily $m = (u-1, q)$, so the factorization, when it exists, is determined by u . The T -fixed grid is precisely X_m . Write

$$r = \omega(d), \quad H = \langle p \bmod m : p \mid d \rangle \leq (\mathbb{Z}/m\mathbb{Z})^\times. \quad (8.3)$$

The group modulo one is the trivial group.

For integer weights, call the constants *compatible* when

$$p \bmod m \mapsto \bar{a}_p \in \mathbb{F}_\ell^\times \quad (p \mid d) \quad (8.4)$$

extends to a homomorphism $H \rightarrow \mathbb{F}_\ell^\times$. In particular, all these residues must be nonzero. Equivalently, every relation $\prod_{p \mid d} p^{n_p} \equiv 1 \pmod{m}$, with $n_p \in \mathbb{Z}$, must satisfy $\prod_{p \mid d} \bar{a}_p^{n_p} = 1$. Set

$$c(\mathbf{a}) = \begin{cases} \varphi(m)/|H|, & \text{if (8.4) holds,} \\ 0, & \text{otherwise.} \end{cases} \quad (8.5)$$

Theorem 8.1 (Integral cyclic-prime coinvariants). *Assume (8.2) and let all weights be integers. Then*

$$D_q(\mathbb{Z}, \mathbf{a})/(T-1)D_q(\mathbb{Z}, \mathbf{a}) \simeq \mathbb{Z}^{\varphi(q)/\ell} \oplus (\mathbb{Z}/\ell\mathbb{Z})^{c(\mathbf{a})2^{r-1}}. \quad (8.6)$$

Consequently every nonunit nonzero Smith invariant of the complete distribution-plus-symmetry presentation equals ℓ , with the multiplicity in (8.6). The answer is independent of all weights a_p with $p \mid m$.

Corollary 8.2 (No nonzero fixed grid point). *If u has prime order ℓ and $(u-1, q) = 1$, then*

$$D_q/(T-1)D_q \simeq \mathbb{Z}^{\varphi(q)/\ell} \oplus \begin{cases} (\mathbb{Z}/\ell\mathbb{Z})^{2^{\omega(q)-1}}, & a_p \equiv 1 \pmod{\ell} \text{ for every } p \mid q, \\ 0, & \text{otherwise.} \end{cases}$$

For $\ell = 2$ and odd q , this recovers the unanchored reflection coinvariant formula. For $\ell > 2$ it gives odd torsion.

The theorem describes a formal quotient. The equation $T = 1$ is an additional symmetry relation. Ordinary Hurwitz-zeta or polylogarithm evaluation is not asserted to satisfy it without an appropriate invariant projection. In particular, the finite torsion here is not torsion of a complex-valued period and proves no numerical independence statement.

8.2 The finite resolution and its fixed part

We record the algebra needed for a proof, including the role of the prime-order assumption. Order the primes dividing q , and for a subset S put $p_S = \prod_{p \in S} p$. The weighted distribution resolution has terms

$$C_j = \bigoplus_{|S|=j} B[X_{q/p_S}], \quad \partial[S, x] = \sum_{p \in S} (-1)^{\#\{v \in S: v < p\}} \left(\sum_{py=x} [S \setminus \{p\}, y] - a_p [S \setminus \{p\}, x] \right). \quad (8.7)$$

Its augmentation is the quotient in (8.1). The transfers and inclusions in different prime directions commute, which gives $\partial^2 = 0$.

Lemma 8.3 (Exactness over every coefficient ring). *The augmentation of (8.7) is exact. Its degree-zero homology is B -free of rank $\varphi(q)$, and the augmented complex is split exact as a complex of B -modules.*

Proof. Here is the finite filtration argument, to make clear that no division by ℓ or a prime weight is used. Filter $[S, x]$ by the positive integer $p_S \text{den}(x)$, which divides q . The leading part of a prime differential retains only the preimages whose denominator is multiplied by that prime; its weight term lies lower in the filtration. At a fixed augmented denominator, additive primary coordinates identify the leading complex with a tensor product of maps

$$B[U_{p^{a-1}}] \longrightarrow B[U_{p^a}], \quad e_b \longmapsto \sum_{v \mapsto b} e_v.$$

At $a = 1$, U_1 is a singleton and the sum has $p - 1$ terms. At $a > 1$, each fiber has p terms. In making this identification, replace the v -primary coordinate in the S -summand by $(\prod_{p \in S, p \neq v} p)^{-1}$ times that coordinate; this removes the unit twists in the other primary components of a root.

Each displayed map is a split injection: its fibers have disjoint support, and one coefficient-one pivot can be selected in each fiber. Its cokernel is free of rank $\varphi(p^a) - \varphi(p^{a-1})$. The tensor product is therefore acyclic in positive degrees and has free degree-zero homology. Lifting a leading boundary and subtracting it lowers the finite filtration. This proves positive-degree exactness. The same lifting in degree one proves that the image into degree zero is strict for the filtration, so the chosen free complements lift to a basis of the quotient. Their total number is

$$\prod_{p^e \parallel q} \left(1 + \sum_{a=1}^e (\varphi(p^a) - \varphi(p^{a-1})) \right) = \varphi(q).$$

Finally a surjection onto the free quotient splits; its kernel is projective. Inducting upward splits the entire augmented complex. $\square$

Let C_j^{mov} be generated by points which are not fixed by T . It is a subcomplex: if y is fixed, then py is fixed, so a root of a nonfixed point cannot be fixed. Since T has prime order, every nonfixed orbit has exactly ℓ elements, and each term of $C_{\bullet}^{\text{mov}}$ is a sum of regular cyclic permutation modules. Put $K_{\bullet} = C_{\bullet} / C_{\bullet}^{\text{mov}}$.

The fixed symbols in the summand indexed by S form the grid

$$X_{m/p_{S \cap P(m)}}.$$

For $p \mid m$, its fixed differential is the usual weighted distribution differential. For $p \nmid d$, exactly one root remains fixed, namely $p^{-1}x$ in that grid. Thus this differential is

$$V_p - a_p, \quad V_p(e_x) = e_{p^{-1}x}. \quad (8.8)$$

All these operators commute. Applying Lemma 8.3 in the prime directions dividing m shows that $K_\bullet$ is quasi-isomorphic to the operator Koszul complex

$$K(V_p - a_p : p \mid d; , D_m(B, \mathbf{a}|_m)). \quad (8.9)$$

The notation means the tensor construction with one exterior generator for each $p \mid d$, differential given by the indicated commuting endomorphisms. The comparison follows by taking homology first in the bounded m -directions; only degree zero survives.

8.3 A cohomology formula which controls all torsion

For a free abelian group M with $T^\ell = 1$, write $\mathcal{N} = 1 + T + \cdots + T^{\ell-1}$ and define

$$\hat{H}^0(M) = \ker(T - 1)/\mathcal{N}M, \quad \hat{H}^1(M) = \ker \mathcal{N}/(T - 1)M,$$

extended periodically with period two. These are the cohomology groups of the complex with differential $T - 1$ leaving even degree and $\mathcal{N}$ leaving odd degree.

Lemma 8.4 (Torsion and the fixed complex). *For B finite free over $\mathbb{Z}$ and $D = D_q(B, \mathbf{a})$,*

$$\mathrm{Tor}_{\mathbb{Z}}(D/(T - 1)D) = \hat{H}^1(D). \quad (8.10)$$

This group is killed by ℓ . Under (8.2), for $\epsilon \in \{0, 1\}$ one has

$$\hat{H}^\epsilon(D) \simeq \bigoplus_{j \equiv \epsilon \pmod{2}} H_j(K(V_p - \bar{a}_p : p \mid d; , D_m(B/\ell B, \bar{\mathbf{a}}|_m))). \quad (8.11)$$

The isomorphism is one of B -modules; the right side is annihilated by ℓ .

Proof. If $kv = (T - 1)w$ for a nonzero integer k , apply $\mathcal{N}$. Freeness over $\mathbb{Z}$ gives $\mathcal{N}v = 0$. Conversely, if $\mathcal{N}v = 0$, then ℓv belongs to $(T - 1)M$ because $\ell - \mathcal{N}$ is divisible by $T - 1$ in $\mathbb{Z}[T]/(T^\ell - 1)$. This proves both (8.10) and the annihilation assertion.

Apply the periodic cohomology complex to the finite resolution (8.7). Taking horizontal homology replaces it by D in degree zero. A regular orbit has an exact periodic complex: the image of $T - 1$ is the coefficient-sum-zero subgroup, and the image of $\mathcal{N}$ is the constant-vector subgroup. Hence removing $C_\bullet^{\mathrm{mov}}$ does not change total cohomology. Both comparisons are justified by finite horizontal width; each total degree contains only finitely many summands.

On $K_\bullet$ the action is trivial, so the vertical complex alternates between zero and multiplication by ℓ . Since each term of $K_\bullet$ is $\mathbb{Z}$ -free, each adjacent multiplication-by- ℓ block can be replaced by reduction modulo ℓ . The horizontal degree j shifts total degree by $-j$, which gives the parity in (8.11). Finally use the quasi-isomorphism (8.9), also after reduction modulo ℓ . $\square$

8.4 Proof of the scalar formula

Let k be an algebraic closure of $\mathbb{F}_\ell$. Since $\ell \nmid \varphi(m)$, the character projectors for $U(m)$ are defined over k . The conductor normal-form argument applies verbatim in this characteristic: it only divides by unit-group orders, and its edges do not divide by a weight or an Euler factor. For clarity, the character χ has vectors

$$Y_{b,\chi} = \sum_{a \in U(b)} \chi(a) e_{a/b}, \quad \mathrm{cond}(\chi) \mid b \mid m.$$

The prime edges are $Y_{pb,\chi} - a_p Y_{b,\chi}$ when $p \mid b$, and $Y_{pb,\chi} - (a_p - \chi(p)) Y_{b,\chi}$ when $p \nmid b$. Successive elimination leaves exactly one conductor generator per character. Therefore $D_m(k, \bar{\mathbf{a}})$ has one

one-dimensional summand for every character of $U(m)$, independently of its weights. On that summand, V_p acts by $\chi(p)$.

The complex in (8.11) consequently splits into scalar Koszul complexes on

$$(\chi(p) - \bar{a}_p : p \mid d). \quad (8.12)$$

A nonzero scalar makes a Koszul complex contractible. If all these scalars vanish, its differential is zero and its degree- j homology has dimension $\binom{r}{j}$. Thus each surviving character contributes 2^{r-1} to either cohomology parity.

The prescribed values in (8.12) occur if and only if (8.4) holds. A character of the subgroup H extends to $U(m)$ over k : one may successively adjoin group elements and take the required roots in the algebraically closed field. All relevant orders are prime to ℓ . The set of extensions is a torsor under the character group of $U(m)/H$, so its cardinality is $\varphi(m)/|H|$. This proves the torsion multiplicity in (8.6); extension to k preserves the dimension over $\mathbb{F}_\ell$.

For the free rank, rational coinvariants are an exact functor because averaging over a finite group is possible over $\mathbb{Q}$. A permutation set with b points and f fixed points has $(b + (\ell - 1)f)/\ell$ orbits. Its alternating count in the resolution gives

$$\text{rank}_{\mathbb{Z}} D/(T-1)D = \frac{1}{\ell} \sum_{S \subseteq P(q)} (-1)^{|S|} \left(\frac{q}{p_S} + (\ell - 1) \frac{m}{p_{S \cap P(m)}} \right) = \frac{\varphi(q)}{\ell}.$$

The second alternating sum vanishes because $r = \omega(d) > 0$. The classification of finitely generated abelian groups and Lemma 8.4 complete the proof of Theorem 8.1.

8.5 All finite integral jets

Let $B_L = \mathbb{Z}[t]/(t^L)$ with $L \geq 1$, and take arbitrary $a_p(t) \in B_L$. Test compatibility using the constant terms $\bar{a}_p(0)$. If compatibility fails, put $c = 0$ and $\nu = 0$. If it holds, let $c = \varphi(m)/|H|$ and define

$$\nu = \min_{p \mid d} \text{ord}_t(\bar{a}_p(t) - \bar{a}_p(0)) \quad \text{in } \mathbb{F}_\ell[t]/(t^L), \quad \text{ord}_t(0) = L. \quad (8.13)$$

In the compatible case $1 \leq \nu \leq L$.

Theorem 8.5 (Cyclic-prime integral jet formula). *Under (8.2), the underlying abelian group is*

$$D_q(B_L, \mathbf{a})/(T-1)D_q(B_L, \mathbf{a}) \simeq \mathbb{Z}^{L\varphi(q)/\ell} \oplus (\mathbb{Z}/\ell\mathbb{Z})^{c2^{r-1}\nu}. \quad (8.14)$$

More precisely, its integer torsion, as a B_L -module, is

$$(B_L/(\ell, t^\nu))^{c2^{r-1}}. \quad (8.15)$$

The free summand in (8.14) is only asserted to be free over $\mathbb{Z}$, not over B_L .

Proof. Over $k[t]/(t^L)$, the same conductor projectors split the fixed module into free rank-one character blocks. A block whose constants do not match has a unit parameter and is contractible. For each of the c matching characters, all parameters are the same list

$$-(\bar{a}_p(t) - \bar{a}_p(0)), \quad p \mid d.$$

In the principal ideal ring $k[t]/(t^L)$, invertible operations on this list transform it into $(t^\nu, 0, \dots, 0)$, interpreting t^L as zero. Both the kernel and the cokernel of multiplication by t^ν are isomorphic to $k[t]/(t^\nu)$. Tensoring with the $r - 1$ remaining zero-differential exterior factors gives

$$H_j \simeq (k[t]/(t^\nu))^{(r)}_j$$

in each matching block. Lemma 8.4 gives the claimed multiplicity.

For the B_L -module assertion no splitting-field ambiguity is needed. The compatible character of H takes values in $\mathbb{F}_\ell$, so its averaging idempotent is defined over $\mathbb{F}_\ell[t]/(t^L)$. Its image in the fixed distribution module is a projective module of rank c over this local ring, hence is free of rank c . On that image every V_p is the prescribed scalar. The other character blocks are contractible after the faithfully flat splitting extension and therefore have zero homology already before that extension. The preceding scalar Koszul calculation can consequently be performed over $\mathbb{F}_\ell[t]/(t^L)$ itself, proving (8.15). Finally the rational orbit count in the scalar proof is multiplied by L , giving the free rank. $\square$

8.6 Explicit examples and boundaries of the result

Level 91, a genuinely odd torsion group. Take $q = 7 \cdot 13$, $\ell = 3$, and $u = 16$. Then u has order three and $(u - 1, 91) = 1$. For $a_7 = a_{13} = 1$,

$$D_{91}/(T - 1)D_{91} \simeq \mathbb{Z}^{24} \oplus (\mathbb{Z}/3)^2.$$

If either of these weights is not one modulo three, the torsion vanishes. At jet length $L = 4$, choose $a_7(t) = 1 + t^2$ and $a_{13}(t) = 1 + 3t + t^3$. Their contact order is $\nu = 2$, so the quotient is $\mathbb{Z}^{96} \oplus (\mathbb{Z}/3)^4$.

Level 455, a compatibility condition involving two primes. Take $q = 5 \cdot 7 \cdot 13$, again with $\ell = 3$ and $u = 16$. Here $m = 5$, $d = 91$, and $H = \langle 2, 3 \rangle = U(5)$ has order four. Because $3 = 2^{-1}$ in $U(5)$ and every nonzero element of $\mathbb{F}_3$ is its own inverse, compatibility is precisely

$$a_7 \equiv a_{13} \not\equiv 0 \pmod{3}.$$

Thus the quotient is $\mathbb{Z}^{96} \oplus (\mathbb{Z}/3)^2$ when the two residues are both one or both two, and is $\mathbb{Z}^{96}$ otherwise. The value of a_5 is irrelevant. The case where both residues are two shows why the criterion cannot be replaced by an independent requirement $a_p \equiv 1$ when a nontrivial fixed denominator remains.

The hypotheses exclude ramified local actions. At $q = 9$, $\ell = 3$, $u = 4$, the fixed denominator is three, but it is not relatively prime to its complementary factor. The five grid orbits give generators $e_0, e_{1/3}, e_{2/3}, A, B$, where A and B correspond to the two orbits of size three. The three raw relations are

$$(1 - a)e_0 + e_{1/3} + e_{2/3} = 0, \quad 3A - ae_{1/3} = 0, \quad 3B - ae_{2/3} = 0.$$

At $a = 0, 1, 2$ these give, respectively,

$$\mathbb{Z}^2 \oplus (\mathbb{Z}/3)^2, \quad \mathbb{Z}^2 \oplus \mathbb{Z}/3, \quad \mathbb{Z}^2.$$

These exact presentations explain why the fixed-grid hypotheses must remain part of the statement. They are not counterexamples to a claim in the inspected manuscript, which did not assert the cyclic-prime formula.

8.7 Verification, integration, and remaining questions

The accompanying program constructs the complete defining rows directly on X_q , identifies points in the orbits of u , and computes the Smith normal form over $\mathbb{Z}$. It imports no normal-form code from either incoming integral-distribution package. Its frozen suite checks scalar weights

for symmetry primes 2, 3, 5, prime-power and composite levels through 455, incompatible weight assignments, and nonconstant integral jets through length four. The checks verify every nonzero invariant, not merely ranks modulo ℓ . They provide independent arithmetic regression evidence; the all-level proofs are the arguments above.

The preceding exact results resolve the proposed prime-order subgroup problem under (8.2). They do not settle it for an arbitrary unit of prime order. Three concrete next problems remain. First, compute the fixed complex for a local action of order ℓ on an ℓ -power denominator, beginning with the displayed level-nine presentation. Second, remove $\ell \nmid \varphi(m)$: the operator Koszul model still applies, but its modular unit-group action need not split into character lines. Third, compute the full module over a multivariate deformation ring, where the scalar Koszul parameters generate a nonprincipal ideal and a single contact order no longer determines its homology. The first two problems concern new stabilizer or modular representation structure; the third concerns the syzygies of an explicit ideal.

9 Resolvent order and a complete shape criterion for index two

The first-order index-two turning point and the complete index-three phase diagram have already been proved in the incoming boundary and global-phase reports [15, 16]. They must not be presented as new conjectures here. The results below add a finite-parameter comparison valid at every index, and a shape classification valid at every derivative order at index two. They also distinguish the second and third derivative orders rigorously: the second-order lower zero still has an interior minimum, whereas both third-order zeros decrease throughout the deformation interval.

9.1 The spectral family and inherited analytic facts

For integers $n \geq 0$, $k \geq 1$, write

$$\begin{aligned} P_k(t) &= \prod_{j=1}^k (1 + t/j) = \sum_{i=0}^k e_i(k) t^i, \\ B_{n,k}(x) &= n! \sum_{i=0}^{\min(n,k)} e_i(k) \frac{x^{n-i}}{(n-i)!}, \\ f_{n,k}(a) &= k! a^{-k-1} B_{n,k}(-\log a), \end{aligned} \tag{9.1}$$

$$F_{n,k}^\rho(a) = \sum_{m \geq 0} \rho^m f_{n,k}(a + m), \quad 0 \leq \rho \leq 1. \tag{9.2}$$

At $\rho = 1$ this is $(-1)^{n+k} \gamma_n^{(k)}(a)$ in the Laurent convention of the manuscript. At $\rho < 1$ it is the same signed parameter derivative of the convergent Lerch coefficient. The classical all-order parameter-derivative identities are established in Coffey's work [13]. The defining series converges absolutely even at $\rho = 1$, since $f_{n,k}(x) = O(x^{-k-1}(1 + \log^n x))$ at infinity. Its a derivatives converge locally uniformly, and

$$(F_{n,k}^\rho)' = -F_{n,k+1}^\rho. \tag{9.3}$$

The Appell–Laplace representation in the manuscript is

$$F_{n,k}^\rho(a) = \int_0^\infty \frac{x^k e^{-ax}}{1 - \rho e^{-x}} Q_n(\log x) dx, \quad \frac{e^{xt}}{\Gamma(1+t)} = \sum_{n \geq 0} Q_n(x) \frac{t^n}{n!}. \tag{9.4}$$

The reciprocal-gamma product makes Q_n strictly real-rooted, and Laplace variation diminution bounds the positive zero count by n , including multiplicities. We retain that established result

as the analytic zero-count input. We shall also use the elementary endpoint signs: $F_{n,k}^\rho > 0$ near $a = 0$ and $\text{sgn } F_{n,k}^\rho = (-1)^n$ for all sufficiently large a . All positive zeros lie in a common compact interval at fixed n, k as ρ ranges over $[0, 1]$. The latter fact follows because the first spectral term diverges positively at zero while the other terms are uniformly bounded there, and all spectral terms have their eventual common sign past the last elementary zero.

9.2 A finite-parameter resolvent comparison

Theorem 9.1 (Resolvent identity and the last positive zero). *Fix $n \geq 0$ and $k \geq 1$. For $0 \leq \rho_1 < \rho_2 \leq 1$ and $a > 0$,*

$$F_{n,k}^{\rho_2}(a) = F_{n,k}^{\rho_1}(a) + (\rho_2 - \rho_1) \sum_{j \geq 1} \rho_2^{j-1} F_{n,k}^{\rho_1}(a+j). \quad (9.5)$$

The series is absolutely convergent, including when $\rho_2 = 1$. If R_1 is the largest positive zero at ρ_1 , then $F_{n,k}^{\rho_2}$ has no zero in $[R_1, \infty)$. Thus, whenever both largest positive zeros exist,

$$R(\rho_2) < R(\rho_1). \quad (9.6)$$

Neither saturation nor simplicity is required. If the function has no positive zero at ρ_1 , it has none at any larger parameter either.

On every interval where the largest zero is simple it is differentiable and satisfies $R'(\rho) < 0$ for $0 < \rho < 1$.

Proof. Since $\rho_1 < 1$, the double series on the right of (9.5) is absolutely convergent even at $\rho_2 = 1$: the coefficient of $|f_{n,k}(a+r)|$, after absolute values and collection by $r = j+m$, is at most $(1 - \rho_1)^{-1}$. The coefficient identity

$$(\rho_2 - \rho_1) \sum_{j=1}^r \rho_2^{j-1} \rho_1^{r-j} = \rho_2^r - \rho_1^r$$

then proves the resolvent formula term by term.

Put $\sigma = (-1)^n$. On (R_1, ∞) the first function has strict sign σ , because there are no more zeros. At every $a \geq R_1$ all shifted values in the sum therefore have that same strict sign. The unshifted value is either of sign σ or zero. Consequently $\sigma F_{n,k}^{\rho_2}(a) > 0$ throughout this closed ray. If no zero exists at ρ_1 , the function has a fixed strict sign on the whole positive axis and the same argument applies everywhere.

Differentiating the resolvent, or directly collecting the absolutely convergent interior series, gives

$$\partial_\rho F_{n,k}^\rho(a) = \sum_{j \geq 1} \rho^{j-1} F_{n,k}^\rho(a+j), \quad 0 \leq \rho < 1. \quad (9.7)$$

At a largest simple zero both $\partial_a F$ and every shifted value have sign σ . Implicit differentiation yields $R' = -F_\rho / F_a < 0$. $\square$

Remark 9.2 (Scope of the finite comparison). The largest zero can disappear in a multiple-zero event. The theorem does not assert its continuity through such an event, nor does it classify all possible inner folds. Its finite inequality applies whenever the two largest zeros being compared exist. In particular it gives strict decrease on $[0, 1]$ for every largest-root family already known to remain simple there.

For later use, all interior parameter derivatives have a similar exact formula. For $r \geq 1$,

$$\partial_\rho^r F_{n,k}^\rho(a) = r! \sum_{j \geq r} \binom{j-1}{r-1} \rho^{j-r} F_{n,k}^\rho(a+j), \quad 0 \leq \rho < 1. \quad (9.8)$$

This follows either from (9.4) and the geometric series or by collecting coefficients. At a largest zero its sign is $(-1)^n$ for every $r \geq 1$. These are partial derivatives of F ; the statement does not say that every derivative of the zero location has a fixed sign.

9.3 Two coefficient sign changes allow only one turning point

Put

$$H_k = \sum_{j=1}^k \frac{1}{j}, \quad s_k = \left(\sum_{j=1}^k \frac{1}{j^2} \right)^{1/2}, \quad A_k = e^{H_k - s_k}, \quad B_k = e^{H_k + s_k}. \quad (9.9)$$

At index two the elementary kernel is

$$f_{2,k}(a) = k! a^{-k-1} ((H_k - \log a)^2 - s_k^2). \quad (9.10)$$

Its sign pattern on the positive axis is $+, -, +$, with zeros A_k, B_k . For a fixed a , the nonzero coefficients $f_{2,k}(a + m)$ therefore have at most two sign changes as m increases.

We recall the elementary power-series Descartes principle. A nonzero real power series with s coefficient sign changes has at most s zeros in $(0, 1)$ counted with multiplicity. For completeness, choose a real cutoff c between the coefficient indices at the first sign change. The coefficients of $(\rho \partial_\rho - c)h$ have one fewer sign change, since the first sign block is reversed. Its transform is $\rho^{c+1}(\rho^{-c}h)'$. Induction and Rolle's theorem with multiplicity prove the assertion, starting with a sign-definite series. All differentiations are justified on compact subintervals of $(0, 1)$. In particular, a series whose nonzero coefficients are negative and then positive has positive derivative at every interior zero: at a separating cutoff c , every nonzero term of $\sum_m (m - c)c_m \rho^m$ is positive.

The incoming index-two theorem gives exactly two simple positive zeros of $F_{2,k}^\rho$ for every $k \geq 1$ and $0 \leq \rho \leq 1$. We denote them by $a_k(\rho) < b_k(\rho)$. A short reconstruction is useful. At $k = 1$ the exact rational evaluation in the present package gives

$$-0.07105864218565829478 < F_{2,1}^1(13/10) < -0.07105864218565829477.$$

At $a = 13/10$ the spectral coefficients are negative and then positive. The single-crossing coefficient argument keeps their sum negative for all smaller parameters, including the elementary endpoint. Positive signs near zero and infinity give two roots, and the global bound makes both simple. Rolle's theorem between the roots, plus a stationary point after the last one because F tends to zero at infinity, preserves two distinct roots at every higher derivative order. The same bound makes them complete and simple.

Theorem 9.3 (All-order shape dichotomy at index two). *For each integer $k \geq 1$ the upper branch b_k is strictly decreasing on $[0, 1]$. The lower branch a_k has at most one stationary point in $(0, 1)$. Any such point ρ_* is a nondegenerate minimum:*

$$a'_k(\rho_*) = 0, \quad a''_k(\rho_*) > 0.$$

It follows that exactly one of the following occurs:

1. $a'_k(\rho) < 0$ for all $0 < \rho < 1$;
2. there is one $\rho_* \in (0, 1)$, with $a'_k < 0$ before it and $a'_k > 0$ after it.

In both cases the branch initially decreases. More explicitly, if $d_k = \log(1 + A_k^{-1})$, then

$$a'_k(0) = -\frac{A_k^{k+2}}{2s_k(A_k + 1)^{k+1}} d_k(2s_k - d_k) < 0. \quad (9.11)$$

Proof. Simplicity and compactness give continuous branches on $[0, 1]$, analytic for $0 \leq \rho < 1$. The upper assertion follows immediately from Theorem 9.1.

Consider a stationary point of the lower branch and fix its location $a = a_k(\rho_*)$. If $a \geq B_k$, every spectral coefficient is nonnegative and some are positive, so no zero is possible. If $A_k \leq a < B_k$, the coefficients are negative and then positive, after ignoring zeros. Their derivative is strictly positive at a zero, so $F_\rho = 0$ is impossible there. Hence $a < A_k$.

Now $h(\rho) = F_{2,k}^\rho(a)$ has positive constant term and at most two coefficient sign changes. At the stationary point, $h(\rho_*) = h'(\rho_*) = 0$. Descartes' bound forces exact multiplicity two and excludes every other parameter zero. Since $h(0) > 0$, the double contact has $h''(\rho_*) > 0$. The lower root has $F_a < 0$. Twice differentiating the implicit equation at the stationary point therefore gives

$$a_k''(\rho_*) = -\frac{F_{\rho\rho}}{F_a} > 0.$$

Every stationary point is consequently a strict minimum, with derivative negative immediately before it and positive immediately after it. There can be at most one: two would force a transition from positive derivative to negative derivative at an intervening zero of the derivative, contradicting the just-proved local behavior.

At $\rho = 0$, implicit differentiation yields $a_k'(0) = -f_{2,k}(A_k + 1)/f_{2,k}'(A_k)$. Here $f_{2,k}'(A_k) = -2s_k k! A_k^{-k-2}$, which gives (9.11). Its strict sign follows from $A_k + 1 < B_k$: indeed $B_k - A_k = 2e^{H_k} \sinh s_k \geq 2e \sinh 1 > 1$. This also gives $0 < d_k < 2s_k$ and completes the dichotomy. $\square$

9.4 The entire shape is determined by one endpoint sign

For $k \geq 2$, the series for F and F_ρ converge absolutely and locally uniformly in a at $\rho = 1$. Thus both zero branches are C^1 on $[0, 1]$. Put $r_k = a_k(1)$ and define

$$C_k(a) = kF_{2,k-1}^1(a) + 2F_{1,k-1}^1(a). \quad (9.12)$$

Theorem 9.4 (Endpoint criterion). *For every integer $k \geq 2$,*

$$a_k'(1) = \frac{C_k(r_k)}{F_{2,k+1}^1(r_k)}. \quad (9.13)$$

The denominator is strictly positive. The lower branch has its unique interior minimum if and only if $C_k(r_k) > 0$. If $C_k(r_k) \leq 0$, then $a_k'(\rho) < 0$ for every $0 < \rho < 1$.

Proof. The elementary generating identity

$$(a + m) f_{n,k}(a + m) = k f_{n,k-1}(a + m) + n f_{n-1,k-1}(a + m)$$

follows by differentiating $(k + t)(1 + t)_{k-1}(a + m)^{-k-t}$ in t exactly n times at zero. Summing it gives, for $k \geq 2$,

$$\partial_\rho F_{n,k}^1(a) = k F_{n,k-1}^1(a) + n F_{n-1,k-1}^1(a) - a F_{n,k}^1(a). \quad (9.14)$$

Use $n = 2$, $F_{2,k}^1(r_k) = 0$ and $F_a = -F_{2,k+1}^1$ to obtain (9.13). The denominator is positive because $F_a < 0$ at the lower simple root.

If this velocity is positive, the negative initial velocity and continuity force a stationary point; the dichotomy gives precisely one. If it is negative, an interior minimum is impossible, since after the only possible minimum the derivative would remain positive all the way to the endpoint.

We include the zero-velocity case to make the criterion complete. Suppose $a_k'(1) = 0$. Set $r = r_k$ and $h(\rho) = F_{2,k}^\rho(r)$. Then $h(1) = h'(1) = 0$. If $r \geq A_k$, the single-crossing argument would instead give $h'(1) > 0$; the sums with the additional factor m are absolutely convergent, so the

same strict weighted-coefficient proof applies at the endpoint. Hence $r < A_k$, and the nonzero coefficients of h have the pattern $+, -, +$.

Choose c between the last negative coefficient index and the first following positive index, and put $p = (\rho \partial_\rho - c)h$. Its nonzero coefficients have the single pattern $-, +$ and $p(1) = 0$. The weighted single-crossing inequality implies $p(\rho) < 0$ for all $0 < \rho < 1$: an appropriate power of ρ times p is strictly increasing to zero at one. Therefore

$$(\rho^{-c}h)' = \rho^{-c-1}p < 0, \quad \lim_{\rho \uparrow 1} \rho^{-c}h(\rho) = 0,$$

and $h(\rho) > 0$ on $(0, 1)$. If the lower branch had an interior minimum, it would increase thereafter toward r ; sufficiently near one, $a_k(\rho) < r < b_k(\rho)$ and hence $h(\rho) < 0$, a contradiction. The zero-velocity case also has no interior minimum. The dichotomy then makes its derivative strictly negative at every interior point. $\square$

The first derivative order can also be recovered directly from these arguments. In (9.4), at $n = 2, k = 1$, the integrand of F_ρ is

$$\frac{x e^{-(a+1)x}}{(1 - \rho e^{-x})^2} ((\log x + \gamma)^2 - \zeta(2)).$$

Its negative part is supported in a fixed compact interval away from zero and has bounded integral, uniformly for a in a fixed compact positive interval. Its positive part near zero increases to a function asymptotic to $x^{-1} \log^2 x$, which has divergent integral. Hence $F_\rho \rightarrow +\infty$ as $\rho \uparrow 1$, uniformly along the compact lower-root branch. Since F_a stays negative and away from zero near its simple endpoint, $a'_1(\rho) > 0$ near one. Together with the initial negative derivative and Theorem 9.3, this reproves the incoming first-order minimum theorem without assuming a numerical minimum scan.

9.5 The second-order reversal and third-order decrease

Theorem 9.5 (A rigorously separated pair of derivative orders). *The second-order lower branch a_2 has exactly one nondegenerate interior minimum. The third-order lower branch a_3 decreases strictly throughout $[0, 1]$, and $a'_3(\rho) < 0$ at every interior parameter. The upper branch decreases at both orders. Consequently $k = 3$ is the smallest individual derivative order for which both index-two branches decrease throughout the interval; the incoming first-order theorem and the present second-order theorem exclude $k = 1, 2$.*

The following are rigorous rational enclosures:

$$\begin{aligned} 1.29982837988436 &< r_2 < 1.29982837988437, \\ 1.77628448580819 &< r_3 < 1.77628448580820, \\ 0.02581658837068013 &< a'_2(1) < 0.02581658837071208, \\ -0.24231787997240310 &< a'_3(1) < -0.24231787997233853. \end{aligned}$$

Proof. Exact arithmetic gives positive and negative endpoint signs, respectively, at the ends of each displayed root interval. The two-zero theorem makes each positive-to-negative crossing the lower simple zero. On the entire root intervals, the same arithmetic encloses

$$\begin{aligned} -2.158373629377 &< F_{2,2,a}^1 < -2.158373629376, \\ 0.0557218435397 &< C_2 < 0.0557218435399, \\ -1.066689311681 &< F_{2,3,a}^1 < -1.066689311680, \\ -0.2584778925956 &< C_3 < -0.2584778925955. \end{aligned}$$

The full unrounded intervals give the displayed velocity enclosures. The signs are therefore valid at the unknown exact roots, and Theorem 9.4 proves the asserted global shapes. $\square$

A separate, non-interval Laplace-integral diagnostic locates the new minimum at

$$\rho_* \approx 0.9998163851082947468, \quad a_2(\rho_*) \approx 1.299826525290189674.$$

The diagnostic gives $a_2''(\rho_*) \approx 86.8818201681$. These decimals are not claimed to be certified minimum enclosures. The existence, uniqueness and nondegeneracy of the minimum follow from the theorems and the exact endpoint signs, not from this numerical solve. The narrow upturn also explains why a coarse parameter grid can miss the second-order failure of monotonicity.

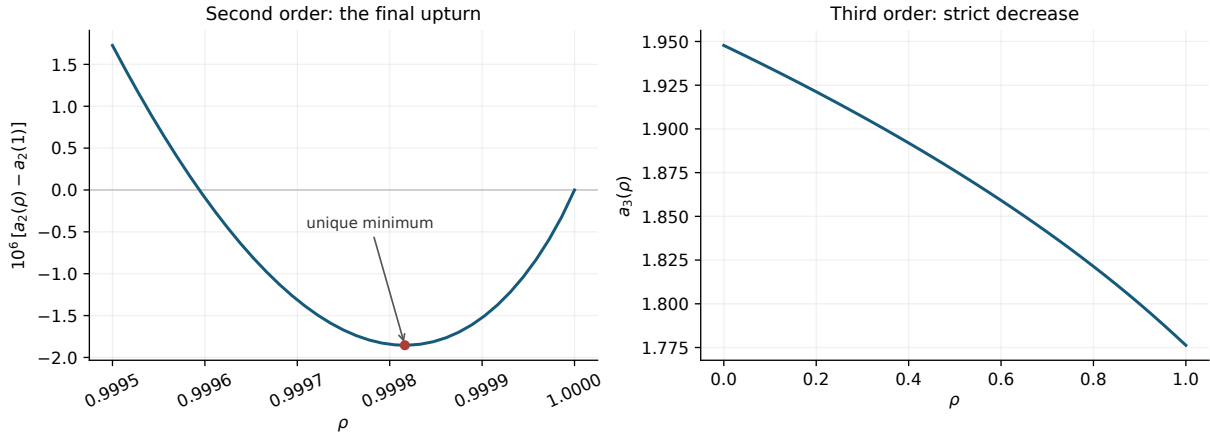


Figure 2: Numerical diagnostics of the proved shapes. Left: the final second-order upturn, on a scale of one million times the difference from the endpoint root. Right: the third-order lower branch. The red minimum marker is the independent Laplace-integral diagnostic. All 79 plotted points and their spectral-tail majorants are retained in `results/lerch/shape_plot_points.csv`; arithmetic rounding is not enclosed. The existence and shape theorems rely on the analytic proofs and exact endpoint certificates.

The last statement of Theorem 9.5 concerns the smallest individual order. A proof that the same property holds for *every* $k \geq 3$ remains a separate question. The inherited eventual-monotonicity theorem proves it for all sufficiently large k ; the endpoint criterion reduces the sharp remaining assertion to $C_k(r_k) \leq 0$ for each $k \geq 3$.

9.6 Exact certificate and its analytic error bound

The proof-bearing computation is `code/lerch/certify_shape.py`, using the exact-rational engine `code/lerch/exact_engine.py`. It uses $N = 32$ initial summands, $p = 8$ Bernoulli corrections, 90 logarithm-series terms and outward rounding to the fixed grid $10^{-70}\mathbb{Z}$. The complete rational data are in `results/lerch/shape_certificates.json`. The engine is copied from the incoming boundary report; the new code adds whole-interval endpoint evaluation and the endpoint shape certificates.

Euler–Maclaurin evaluation of Hurwitz-zeta derivatives with rigorous error bounds also has an established algorithmic literature, notably Johansson [14]. Here is the particular analytic contract used for our rational certificate. If $b = a + N$, the Euler–Maclaurin formula and $f'_{n,k} = -f_{n,k+1}$

give

$$F_{n,k}^1(a) = \sum_{m=0}^{N-1} f_{n,k}(a+m) + f_{n,k-1}(b) + \frac{1}{2}f_{n,k}(b) + \sum_{j=1}^p \frac{B_{2j}}{(2j)!} f_{n,k+2j-1}(b) + R. \quad (9.15)$$

Write $s = k + 2p$ and $B_{n,s}(x) = \sum_{q=0}^n c_q x^q$, where every $c_q \geq 0$. The periodic Bernoulli remainder and an elementary integral give

$$|R| \leq \frac{|B_{2p}|s!}{(2p)!} \sum_{q=0}^n c_q \int_b^\infty x^{-s-1} \log^q x \, dx, \quad \int_b^\infty x^{-s-1} \log^q x \, dx = b^{-s} \sum_{j=0}^q \frac{q!}{(q-j)!} \frac{\log^{q-j} b}{s^{j+1}}. \quad (9.16)$$

All instances use $b > 1$. For an interval of possible a values, natural outward interval arithmetic is applied to every finite term of (9.15); evaluating the positive integral bound from the smaller b gives a valid bound for the whole interval. This is what certifies C_k at the actual root rather than only at a rounded root.

The logarithm is range-reduced to $1 \leq y \leq 2$, then evaluated by

$$\log y = 2 \sum_{j=0}^{M-1} \frac{u^{2j+1}}{2j+1} + R_M, \quad u = \frac{y-1}{y+1}, \quad 0 \leq R_M \leq \frac{2u^{2M+1}}{(2M+1)(1-u^2)}.$$

Every number in the acceptance calculation is rational. The JSON file records rational interval endpoints and their outward decimal displays; no floating-point sign enters the theorem.

9.7 Further questions and editorial consequences

1. Prove or refute $C_k(r_k) < 0$ for every $k \geq 3$. This is a concrete all-order inequality in classical Stieltjes derivatives, exactly equivalent to the proposed sharp persistent monotonicity threshold once the possible equality case is included.
2. Derive a recurrence or comparison inequality in k for $C_k(r_k)$. Such a comparison would turn the third-order certificate into an all-higher-order theorem, avoiding an unlimited list of finite checks.
3. Certify a narrow two-dimensional box for the second-order minimum. The parameter $\eta = -\log \rho$ and the existing Abel singularity subtraction are suitable coordinates; a raw geometric truncation converges slowly at the observed parameter.
4. Extend the resolvent comparison from the largest zero to selected inner zeros using additional sign information on the shifted values. The formula already specifies exactly which weighted tail controls a root's velocity.
5. Determine which discontinuities of the largest-root function can occur at higher indices. Strict finite-parameter order excludes an upward jump; classifying possible losses of the outermost pair requires information beyond the order theorem.

No inspected proof of the existing first-order shape or the eventual monotonicity theorem is contradicted. The editorial correction is to keep three statements distinct: first-order nonmonotonicity, the new second-order nonmonotonicity, and third-order strict decrease. A claim that the eventual threshold is exactly three still needs the all- k inequality in the first question. Likewise, the earlier formal-reductions report's open first-order minimum question has already been settled by the incoming boundary report and should be cross-referenced accordingly.

10 A certified obstruction in the weight-seven reduction search

The weight-seven identity in the pinned manuscript [1] remains conjectural. Here $S_6 = \sum_{n \geq 0} (-1)^n H_n / (2n+1)^6$ and $H_0 = 0$. Its equivalent short form is

$$S_6 = -\frac{12334}{527}g_{6,1} - \frac{384}{31}g_{5,2} - \frac{2136}{527}g_{4,3} + \frac{3179\pi^7}{5713920} - \frac{801}{2108}\beta(4)\zeta(3) - 2\beta(6)\log 2. \quad (10.1)$$

The earlier formal obstruction involves products of two single values. We tested a substantially larger finite family, allowing double-shuffle rows through depth four, convergent distribution and its lifts, products of the already proved S_2 and S_4 relations, and a specified family of Cayley identities whose expansions stay within depth four. An exact integer functional separates the target from this enlarged span. This is a result about the stated formal presentation; it neither disproves the numerical conjecture nor excludes identities outside that presentation.

Use the forms

$$z = dt/t, \quad c = dt/(1-t), \quad A = i dt/(1-it), \quad B = -dt/(1+t), \quad \bar{A} = -i dt/(1+it),$$

and outer-letter-first iterated integrals on $[0, 1]$. Admissible words have first letter different from c and last letter different from z . Their depth is the number of letters different from z . Let $\mathcal{W}$ be the rational space on imaginary admissible words of weight seven and depth at most four, with conjugates identified with opposite signs. It has dimension 2546. More explicitly, the dimensions at exact depths one through four are 1, 35, 390, 2120; they count formal words, not independent periods.

The interval-reversing Cayley substitution $\phi(t) = (1-t)/(1+t)$ acts by reversal and the letter map

$$\tau z = c - B, \quad \tau c = z + B, \quad \tau A = B - \bar{A}, \quad \tau B = B, \quad \tau \bar{A} = B - A.$$

The displayed letter map is $\tau = -\phi^*$, with the reversed orientation included; it is not the unadjusted pullback. Regularized double shuffle is used with the standard single-divergence convention [8, 9]. It becomes especially convenient on the combinations

$$x = c - B, \quad y = A - \bar{A}, \quad h = A + \bar{A} - B, \\ \tau z = x, \quad \tau x = z, \quad \tau y = y, \quad \tau h = -h, \quad \tau B = B.$$

For an abstract word u on $\{z, x, y, h, B\}$ let $\hat{u}$ denote its expansion in the original alphabet. If its first letter is not x and its last letter is not z , every expanded word and every Cayley-transformed expanded word is admissible. If u has length l and contains n_z copies of z and n_x copies of x , the two expansions have depths at most $l - n_z$ and $l - n_x$. In particular, the $z^3 x^3 y$ permutations supply weight-seven Cayley rows that remain in $\mathcal{W}$ on both sides. This is a useful way to add geometric relations without immediately introducing all depth-seven coordinates.

The certificate fixes the following finite row family. Each row is projected to its imaginary part, nonzero rational multiples are identified, and duplicates are removed.

1. All weight-seven shuffle-minus-stuffle rows whose two input depths sum to at most four. Inputs are admissible, except that one factor may be $\text{Li}_1(1)$, using the standard one-divergence comparison whose divergent word cancels.
2. All convergent level-two-to-level-four distribution rows, and their products with admissible words, when total weight is seven and the sum of depths is at most four. Products are expanded by stuffle. In relative-color notation, the seed is

$$Z(\mathbf{s}; 2\mathbf{c}) - 2^{|\mathbf{s}|-d} \sum_{\epsilon \in \{0,2\}^d} Z(\mathbf{s}; \mathbf{c} + \epsilon), \quad c_j \in \{0, 1\}.$$

3. The imaginary S_2 and S_4 relations already proved in the manuscript, multiplied by real parts of all admissible words of the complementary weight, with total depth at most four.
4. For every $1 \leq l \leq 7$, every abstract word u as above with $\min(n_z, n_x) \geq 1$, take its convergent Cayley row and multiply it by each admissible word of weight $7 - l$, provided $l - \min(n_z, n_x)$ plus the multiplier's depth is at most four. The empty multiplier is allowed when $l = 7$.

This prescription gives 5131 distinct nonzero rows: 3460 in the first family, 1394 newly added distribution rows, 40 newly added S_2/S_4 lifts, and 237 newly added balanced Cayley lifts. “Newly added” here concerns duplicate removal in this finite row list, not claims of new mathematical identities.

Let $T \in \mathcal{W}$ represent 128588 times the left side minus the right side of (10.1). The normalization uses $S_6 = g_{6,1} + \text{Im Li}_{6,1}(i, -1)$ and $\beta(7) = 61\pi^7/184320$. Before projecting and expanding the two products, it is

$$\begin{aligned} T = \text{Im}\{ & 128588 \text{Li}_{6,1}(i, -1) + 3138084 \text{Li}_{6,1}(i, 1) + 1592832 \text{Li}_{5,2}(i, 1) \\ & + 521184 \text{Li}_{4,3}(i, 1) - 216172 \text{Li}_7(i) + 48861 \text{Li}_4(i) \text{Li}_3(1) \\ & - 257176 \text{Li}_6(i) \text{Li}_1(-1)\}. \end{aligned}$$

Here T denotes the corresponding formal vector, not its as-yet unproved evaluation as a number.

Theorem 10.1 (A finite enlarged-span obstruction). *The target T does not belong to the rational span of the 5131 rows specified above.*

Proof. The supplementary file `S6_depth_bounded_separator.json` contains an integer functional ℓ on $\mathcal{W}$. The independent program `verify_s6_separator.py` reconstructs every prescribed row from its descriptor, using placement shuffles and stack-enumerated index merges. It checks admissibility, weight seven, and depth at most four. For Cayley rows it performs the raw five-letter substitution, independently of the diagonalized-letter shortcut used during discovery. It also independently re-enumerates the full prescribed family and verifies that its normalized row set equals the stored descriptor list. It then verifies, using only integers, that every row R has $\ell(R) = 0$, whereas $\ell(T) \neq 0$. The explicit nonzero pairing is recorded in the receipt. Linear algebra now proves the assertion. The replay uses no rank assumption, modular arithmetic, numerical polylogarithm, or integer-relation search. $\square$

This certificate narrows one proof-search strategy. The family is still smaller than all relations obtainable through arbitrary higher-depth elimination. Even within a depth-bounded ansatz, other convergent linear combinations of Cayley words need not be generated by these balanced seeds. Thus the theorem does not prove that every possible proof must pass through depth five. Its concrete recommendation is to enlarge this specified row family or change its coordinate system, rather than repeat the same search at higher numerical precision.

11 Verification, editorial reconciliation, and integration

11.1 What the computations establish

The analytic theorems are proved in the text. Exact computation enters at three clearly identified places: enclosure of the Gaussian maximum, endpoint signs for the two Lerch branch-shape decisions, and separation of a finite formal relation span. The distribution examples and inverse coefficient checks are independent consistency tests of general proofs.

Check	Construction	Logical role
Gaussian maximum	Outward rational intervals, finite Euler transforms, analytic derivative tails	Certifies the printed root and maximum intervals once strict log-concavity is proved.
Inverse expansion	Exact symbolic generalized series, with exponents encoded by rational bases	Substitution cancels every coefficient below exponent two; the all-order theorem has an analytic proof.
Cyclic quotients	Exact Smith forms of the original orbit relation matrices, without a normal-form implementation	52 scalar and finite-jet cases check the theorem; three ramified examples test its boundary.
Lerch branch shapes	Exact Euler–Maclaurin intervals on entire rational root brackets	Certifies the signs deciding the order-two and order-three global branch shapes.
S_6 separation	Integer functional, independent row reconstruction and complete-family enumeration	Proves nonmembership in the stated 5,131-row rational span.
Plots and quadrature	Floating-point evaluation of independent integrals and asymptotic formulas	Diagnostic evidence only; no claimed certificate follows from a plot or an unenclosed decimal.

The nine Gaussian comparison points evaluate the original Euler transform and an independent beta-mixture quadrature. Their largest observed discrepancy is below $6 \cdot 10^{-12}$ with the supplied quadrature settings. This is a diagnostic discrepancy, not a proved bound on the quadrature error. Four inverse examples compare the exact coefficient formula with numerical solutions. The Lerch plot retains its sampled values and analytic omitted-function estimates; floating rounding is explicitly unenclosed. In particular, the diagnostic coordinates of the order-two minimum are not a certified isolating box. Its existence, uniqueness, and nondegeneracy are theorems independently of those coordinates.

The programs and receipts are included in the archive. Running `code/replay_all.py` from the package root regenerates all exact receipts. Optional plotting dependencies are separated from the exact arithmetic dependencies. No network access or numerical integer-relation search is needed for the replay. The distribution checks use SymPy’s exact integer Smith algorithm; the Gaussian, Lerch, and S_6 certificate arithmetic uses the Python standard library.

This is stronger than reporting successful high-precision fitting, but it remains ordinary mathematical proof with auditable programs. Formal verification in a proof assistant is a separate project. For that project, the integer separator is an especially small logical target: the checker need only reconstruct finite vectors and evaluate integer dot products.

11.2 Concrete proposed status corrections

The reviewed material did not reveal a new counterexample to an already established upstream theorem in the portions used here. The useful corrections concern cross-report status and the scope of claims made while integrating results.

Promote the critical constant conjecture. In [2], replace the status of Conjecture `conj:constant` by a theorem, with the proof through Theorems 2.2 and 3.2. Retain the

analytic/Abel convention on $a + b = 1$ and the endpoint atom. The constant is a supremum over the open parameter interval, approached as $a \downarrow 0$; there is no admissible maximizing pair on the line.

Reconcile the first-order Lerch minimum. The formal-reductions report [4] still lists uniqueness of the smaller $n = 2, k = 1$ branch minimum as Conjecture `research:conj:minimum`. The incoming boundary report [15] proves that uniqueness. A combined manuscript should cite the latter result and remove the obsolete open status. The present contribution is the all- k dichotomy, the endpoint criterion, the order-two minimum, and the order-three decreasing case. These should not be described as the first proof of the already settled $k = 1$ case.

Keep the S_4 and S_6 statuses distinct. The current manuscript has a proof chapter for S_4 and an explicitly conjectural S_6 reduction in `chapters/04-cyclotomic-quotients.tex`. The exact separator in Section 10 is a theorem about one finite presentation. It supplies neither a proof nor a disproof of the S_6 evaluation. Its enlarged row family should replace a repeated search in the same smaller ansatz, rather than change the conjecture’s numerical status.

Separate two different Euler constants. The global constant C_* controls $-2g_{a,b}$ over all positive orders. The critical constant $\pi/4 + (\log 2)/2$ controls every normalized Euler tail on $a + b = 1$. The exact $(a, b) = (1, 1)$ calculation in Section 7 proves that normalized errors need not decrease off the critical line. This is a warning against an additional claim, not a counterexample to either theorem stated here or to an identified upstream theorem.

State the cyclic hypotheses where the formula is used. The restrictions $(m, d) = 1$, $(u - 1, d) = 1$, and $\ell \nmid \varphi(m)$ are part of the theorem. They should accompany any extracted formula or summary. The level-nine computation shows why a partially fixed primary factor is a distinct local problem. Likewise, the term “rank” should specify whether it is a free abelian rank, a formal rational dimension, or a rank after reduction modulo a prime. No one of these determines the others without a proof.

11.3 Suggested repository placement

The complete report can first be placed under `Analysis/Polylogarithms/docs/reports/extremal-bounds-cyclic-descent/`. The integration guide supplies a compact addendum and a dependency map. The natural manuscript neighborhoods are:

- Gaussian allocation, the critical Euler theorem, and the secant identity next to the signed-kernel material and the real-order report;
- the cyclic theorem and finite jets next to `08-distribution.tex` and `08-conductor-jets.tex`;
- the Lerch branch theorems next to `09-zero-geometry.tex` and the incoming boundary analysis;
- the finite S_6 obstruction next to `04-cyclotomic-quotients.tex` and the established S_4 proof.

The manuscript need not absorb the entire report at once. Its theorem statements and their proof dependencies can be integrated by topic, while the exact checkers and complete provenance remain with the report. No repository mutation is part of this deliverable.

12 Further research questions and proposed directions

The questions below are intentionally differentiated. Some preserve existing conjectures, some ask for stronger theorems suggested by the present proofs, and some concern a concrete missing computation. No conjecture in this section is used earlier.

12.1 Sharp bounds for the entire Euler scheme

1. Optimize every normalized remainder. For $N \geq 1$, determine

$$M_N = \sup_{a,b>0, a+b \geq 1} 2^N (E_N - g_{a,b}), \quad M = \sup_{N \geq 1} M_N.$$

We have determined $M_1 = C_*$. On the critical line the corresponding all- N supremum is $\pi/4 + (\log 2)/2$. The positive secant identity reduces the general problem to a beta average of divided differences of H_N . The main obstacle is concrete: H_1 is concave on $[1, \infty)$, whereas H_2 is convex near one. A useful next step is to locate the inflection structure of H_N exactly and determine whether its Mellin integral restores monotonicity in the allocation parameter. We make no untested assertion that $M = C_*$.

2. Quantify allocation monotonicity. The covariance formula proves a strict sign but does not give a sharp modulus. Find useful two-sided estimates for

$$-\frac{d}{da} \{-g_{a,w-a}\} = -\text{Cov} \left(J_w(V_a), \log \frac{V_a}{1-V_a} \right),$$

uniformly near $a = 0$ or $a = w$, or as $w \rightarrow 0$ and $w \rightarrow \infty$. The endpoint beta laws concentrate at different rates and have logarithmic score singularities. Deriving a full endpoint expansion would turn the sharp inequalities into effective error estimates for nearly extremal orders. It would also test whether higher derivative signs hold on a restricted range; they do not follow from stochastic ordering alone.

3. Extend the allocation principle to other arguments. For $z = e^{i\theta}$ the positive double integral still supplies an explicit rational imaginary kernel, but its dependence on the allocation variable can change sign and shape with θ . Characterize the angles for which an appropriate signed component is monotone at fixed total order. This would connect the Gaussian theorem to the incoming angular-zero program. In particular, the below-threshold angular uniqueness conjecture from [2] remains open here: failure of a finite signed Hausdorff measure for $a + b < 1$ does not imply failure of the positive double-integral method.

4. Determine the analytic reach of the inverse expansion. Theorem 5.1 gives an asymptotic expansion to every finite order. Does a naturally grouped generalized-power series converge on a nontrivial interval? The infinitely many exponents and their multiplicative collisions make this a distinct question from a finite-variable implicit-function theorem. A second target is a uniform approximation joining the large-order branch to the square-root behavior at the nondegenerate maximum. The latter behavior follows locally from $C''(w_*) < 0$; a global error-controlled transition would require estimates beyond the local theorem.

12.2 Integral distribution and cohomology

5. Include ramified fixed grids. Remove the requirement that each primary factor be either entirely fixed or fixed only at zero. The action $q = 9, u = 4$ gives the first model. A

resolution indexed by both denominator and fixed-point depth should replace the present one-step fixed/moving split. The target is an explicit integral Smith formula, not merely its reduction modulo ℓ . This distinction matters when higher powers of ℓ could appear in more general symmetry groups.

6. Remove semisimplicity on the fixed factor. When $\ell \mid \varphi(m)$, character projectors cannot be obtained by division by the unit-group order. The fixed Koszul complex still gives a useful starting point, but generalized eigenspaces and nilpotent operators replace scalar components. Determine whether the answer can be stated using Fitting ideals of the operators $V_p - a_p$. Small exact Smith computations should guide the statement before a general formula is proposed.

7. Replace one jet parameter by several. For

$$B = \mathbb{Z}[t_1, \dots, t_s] / (t_1^{L_1}, \dots, t_s^{L_s}),$$

the contact ideal generated by the reduced nonconstant weights need not be principal. The proof here already identifies the appropriate Koszul homology. Its length, annihilator, and module structure are the next invariants to compute. A single minimum valuation cannot capture them. Complete-intersection cases and monomial contact ideals are natural initial families with reproducible algebraic tests.

8. Compute composite cyclic and noncyclic symmetries. For a cyclic group of composite order, moving orbits can have several stabilizers. For a noncyclic group, the periodic two-column argument must be replaced by a fuller group resolution. Determine how the prime-order formulas fit together under restriction and transfer. An initial target is a product of two cyclic prime-order groups acting on distinct primary factors. Classical universal norm distributions [12] provide an important comparison framework; the novelty target should be an explicit finite integral formula for the chosen weighted presentation.

12.3 Global Lerch zero geometry

9. Decide the persistent index-two monotonicity threshold. Theorem 9.5 proves that order three is the smallest individual derivative order for which both positive zero branches decrease throughout $0 \leq \rho \leq 1$. It does not prove that every $k \geq 3$ has this property. With the notation of Theorem 9.4, the persistent statement is equivalent to the endpoint inequalities

$$C_k(r_k) \leq 0 \quad \text{for all integers } k \geq 3.$$

This is a precise reduction of the remaining question to one sign per order. A uniform large- k expansion with a rigorous remainder, combined with finitely many exact certificates, is a plausible route. Such a proof should keep the zero-saturation threshold and the branch-monotonicity threshold distinct.

10. Certify the order-two turning point itself. Existence and uniqueness of its nondegenerate minimum are already proved here. Its approximate coordinates are close to the singular endpoint $\rho = 1$, which makes naive finite differences unreliable. Produce a rational isolating rectangle for the simultaneous equations $F_{2,2}^{(\rho)}(a) = 0$ and $\partial_\rho F_{2,2}^{(\rho)}(a) = 0$, with a two-variable interval Newton certificate. This would upgrade the present diagnostic coordinates without changing the proved branch-shape theorem.

11. Extend the stationary-point bound to higher index. For general n , the Appell kernel bounds the number of positive a -zeros by n . The resolvent theorem already controls the largest one without assuming the bound is saturated. What is the optimal number and type of stationary points on the other simple branches? The index-two proof uses three coefficient-sign blocks. Higher index suggests a finite combinatorial constraint, but a sign-change count alone does not determine the type or global order of the stationary points. A useful theorem would combine multiplicity at a stationary parameter with the known sign of F_a on that branch.

12.4 Weight-seven identities and formal verification

12. Enlarge the exact S_6 relation system strategically. The integer separator provides a direct test for any proposed new row in the same ambient word space: if its pairing is nonzero, it crosses the presently certified obstruction. One route is to add convergent linear combinations of Cayley words beyond the balanced seeds. Another is to allow higher-depth coordinates and then eliminate them. A third is to seek a functional identity before evaluation at i . Standard relations and their possible incompleteness are discussed in [8, 9]. The target remains the concrete S_6 identity, not a blanket assertion that one selected relation family is complete.

13. Formalize the finite and analytic interfaces. The separator checker, Smith examples, and outward interval primitives have small trusted mathematical interfaces. Formalizing these first would provide useful infrastructure without immediately formalizing the full special-function library. On the analytic side, the beta covariance lemma and the strict normalized Mellin comparison are reusable standalone lemmas. A good dependency order is positivity and dominated differentiation, then the stochastic comparison, then the special-function endpoint identities. This makes the formalization serve several results rather than duplicate each proof separately.

These questions preserve the scope of the theorems already established. The strongest immediate targets are the persistent index-two endpoint sign, the ramified primary distribution calculation, and an S_6 row with nonzero pairing against the supplied separator. Each has a precise success criterion and a computational tool in the present package.

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